

ABG Interpretation Guide

PulmoLearn | A quick reference for respiratory therapy students

Use this sequence: pH → primary disorder → compensation → oxygenation → final interpretation and clinical context.

Normal adult ABG values

Measurement	Typical range	Meaning
pH	7.35–7.45	Acid-base status
PaCO ₂	35–45 mmHg	Ventilation / respiratory component
HCO ₃ [−]	22–26 mEq/L	Metabolic component
PaO ₂	80–100 mmHg on room air	Oxygenation (age and altitude matter)
SaO ₂	95–100%	Arterial oxygen saturation

The five interpretation decisions

1. Start with pH — Below 7.35 = acidemia; above 7.45 = alkalemia. A normal pH can still occur with compensation or mixed disorders.
2. Find the primary problem — Respiratory: pH and PaCO₂ move in opposite directions. Metabolic: pH and HCO₃[−] move in the same direction.
3. Assess compensation — Ask whether the other system moves in the expected direction and by the expected amount. pH +1: one of PaCO₂/HCO₃[−] abnormal; pH +2: both abnormal. These are screening patterns, not proof of compensation.
4. Evaluate oxygenation — Interpret PaO₂ with FiO₂, oxygen delivery device, age, altitude, clinical findings, and trends.
5. State the full interpretation — Name the primary disorder, compensation pattern (when supported), oxygenation, and clinical significance.

Compensation and the 7.40 midpoint

If pH is 7.35–7.45 and PaCO₂ and HCO₃[−] are abnormal, use 7.40 as an introductory clue: below 7.40 leans acidotic; above 7.40 leans alkalotic. Confirm with expected-compensation calculations and clinical context. Two abnormal components can indicate a mixed disorder.

Respiratory disorders: Acute PaCO₂ change of 10 mmHg → HCO₃[−] shifts about 1 mEq/L (acidosis) or 2 mEq/L (alkalosis); chronic changes are about 3.5–4 up or 4–5 down, respectively. Metabolic acidosis: Winter's formula: expected PaCO₂ ≈ 1.5 × HCO₃[−] + 8 ± 2.

Recognize the four primary disorders

Disorder	Primary pattern	Common examples
Respiratory acidosis	pH ↓ / PaCO ₂ ↑	Hypoventilation, COPD exacerbation, sedation
Respiratory alkalosis	pH ↑ / PaCO ₂ ↓	Hypoxemia, pain, sepsis, hyperventilation
Metabolic acidosis	pH ↓ / HCO ₃ [−] ↓	DKA, lactic acidosis, renal dysfunction, diarrhea
Metabolic alkalosis	pH ↑ / HCO ₃ [−] ↑	Vomiting, diuretics, volume depletion

Oxygenation categories (introductory room-air reference)

PaO ₂	Category	Clinical note
80–100	Typical adult range	Interpret with age, altitude and FiO ₂
60–79	Mild hypoxemia	Assess context and trends
40–59	Moderate hypoxemia	Significant impairment; evaluate promptly
Below 40	Severe hypoxemia	Urgent clinical assessment

Practice ABGs

ABG	Interpretation
pH 7.28 / PaCO ₂ 58 / HCO ₃ ⁻ 26 / PaO ₂ 62	Primary respiratory acidosis; minimal bicarbonate change, with mild hypoxemia on room air.
pH 7.50 / PaCO ₂ 30 / HCO ₃ ⁻ 23 / PaO ₂ 88	Primary respiratory alkalosis with oxygenation in the typical room-air range.
pH 7.36 / PaCO ₂ 55 / HCO ₃ ⁻ 30 / PaO ₂ 68	Respiratory acidosis with renal compensation pattern; confirm expected compensation and history. Mild hypoxemia on room air.

Common pitfalls

Do not assume both abnormal PaCO₂ and HCO₃⁻ always equal compensation. Do not diagnose an acid-base disorder from a single value. Do not interpret PaO₂ without FiO₂. ABGs support, but never replace, bedside assessment.

Continue learning

Five guided learning Steps: pulmolearn.com/abg/case-1.html (start with pH) → [case-2.html](https://pulmolearn.com/abg/case-2.html) → [case-3.html](https://pulmolearn.com/abg/case-3.html) → [case-4.html](https://pulmolearn.com/abg/case-4.html) → [case-5.html](https://pulmolearn.com/abg/case-5.html). Then practice with the ABG Challenge: pulmolearn.com/abg/abg-challenge.html.

Full guide and additional resources: pulmolearn.com/abg/